

Nehal Choudhary

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EDUCATION

University of California - San Diego | San Diego, CA

B.S. Mathematics-Computer Science | B.S. Cognitive Science-Machine Learning

Minor: Economics | **GPA: 3.9** | Expected: June 2027 | Accelerated senior standing (250 units completed)

Relevant Coursework: Data Science in Practice, Modeling & Data Analysis, Probability, Linear Algebra, Econometrics, Data Structures

EXPERIENCE

Machine Learning Intern | Netcracker Technology

July 2025 - Sept 2025

- **Designed and automated evaluation pipelines for an LLM-based analytics system**, running 100+ structured tests to measure precision, recall, consistency, and reliability across prompt versions.
- **Conducted controlled A/B-style experiments on system prompt configurations**, comparing failure modes and performance trade-offs to identify the most reliable configurations.
- **Analyzed knowledge-base integration and JavaScript plugin architectures**, documenting trade-offs that improved retrieval transparency and reduced debugging overhead for downstream teams.

Research on AI Bias in Facial Recognition Models

Aug 2023 - May 2024

- **Analyzed 240+ NIST FRVT samples using regression-style evaluation and statistical hypothesis testing** to quantify demographic error disparities, including precision, recall, and false-positive rate differences.
- **Synthesized quantitative findings into interpretable reliability and fairness insights**, identifying where model performance degraded across demographic groups and communicating implications for real-world deployment.

Director of Curriculum & Operations | CS foreach | UCSD

Jan 2025 - Present

- **Managed operations and officer recruitment for a 10+ person organization**, overseeing onboarding, role coordination, and end-to-end logistics for workshops and organization-wide events.
- **Designed and delivered Python workshops for K-12 partners**, standardizing curriculum materials and incorporating engagement feedback to improve consistency and scalability.

PROJECTS

Spotify Similarity Recommender | Python · FAISS · pandas · Streamlit

- **Designed and evaluated an end-to-end ETL and retrieval pipeline**, ingesting and validating 900+ Spotify audio-feature records to ensure schema consistency and reproducibility for downstream similarity search.
- **Engineered a semantic feature representation and implemented FAISS-based cosine similarity retrieval**, benchmarking nearest-neighbor outputs across multiple queries to assess stability and relevance across tracks.
- **Built an interactive Streamlit dashboard** to visualize similarity relationships and clustering behavior, enabling interpretable analysis and communication of model behavior to non-technical users.

SPY Long/Flat ML Strategy | Python · scikit-learn · Jupyter

- **Developed a walk-forward classification pipeline for financial time series**, including feature engineering, temporal train-test splits, and regularized model selection to mitigate look-ahead bias.
- **Evaluated strategy robustness using risk and stability metrics** (Sharpe ratio, volatility, drawdown) and monitored performance drift across backtesting windows.

TECHNICAL SKILLS

Languages: Python, SQL, Java, HTML/CSS/Javascript

ML & Data: pandas, NumPy, scikit-learn, FAISS; regression, classification, time series, experimentation

Visualization & Tools: Streamlit, Power BI, Git, Jupyter

UPCOMING ENGAGEMENTS

Infant Cognition Lab | ML-based behavioral data analysis (starting Winter 2026)

RoboSub UCSD | Perception / Autonomy team (starting Winter 2026)